

Derivation of the Blackbody Energy Density

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1. Introduction

This document presents the standard derivation of the spectral energy density of blackbody radiation, based on counting electromagnetic modes inside a cubic cavity of volume $V = L^3$ at thermal equilibrium.

2. Counting the Electromagnetic Modes

2.1. Quantization in a Cubic Cavity

Consider a perfectly conducting cube of side L . Electromagnetic standing waves must have nodes at the walls, leading to the condition that each component of the wavevector $\mathbf{k} = (k_x, k_y, k_z)$ must be

$$k_x = \frac{\pi}{L} n_x, \quad k_y = \frac{\pi}{L} n_y, \quad k_z = \frac{\pi}{L} n_z,$$

where n_x, n_y, n_z are positive integers $(1, 2, 3, \dots)$. Thus, all allowed $\mathbf{k}$ -vectors populate a discrete lattice in $\mathbf{k}$ -space, with spacing $\Delta k = \pi/L$ in each direction.

2.2. Relation to the Frequency

For free electromagnetic waves in vacuum, the dispersion relation is $\omega = c k$, where $k = \|\mathbf{k}\| = \sqrt{k_x^2 + k_y^2 + k_z^2}$, and $\omega = 2\pi \nu$, with ν the frequency. Hence,

$$\nu = \frac{c k}{2\pi}.$$

2.3. Counting Modes in k -Space

To find how many modes lie below a certain wave number k , we look at the sphere of radius k in $\mathbf{k}$ -space. Its volume is

$$\frac{4\pi}{3} k^3.$$

Since only $k_x, k_y, k_z > 0$ are allowed (standing-wave boundary conditions), we use one octant of that sphere:

$$\text{Volume in the first octant} = \frac{1}{8} \frac{4\pi}{3} k^3 = \frac{\pi k^3}{6}.$$

Each mode occupies a little “cell” of volume $(\pi/L)^3$ in $\mathbf{k}$ -space, so the *number* of modes up to radius k is approximately

$$N_{\leq k} = \frac{(\text{octant volume in } \mathbf{k}\text{-space})}{\left(\frac{\pi}{L}\right)^3} = \frac{\frac{\pi k^3}{6}}{\frac{\pi^3}{L^3}} = \frac{L^3}{6\pi^2} k^3.$$

However, each $\mathbf{k}$ has *two* independent polarizations (transverse electric and magnetic), so we multiply by 2:

$$N_{\leq k} = 2 \times \frac{L^3}{6\pi^2} k^3 = \frac{L^3}{3\pi^2} k^3.$$

2.4. From k to ν

Because $\nu = \frac{ck}{2\pi}$, we have

$$k = \frac{2\pi\nu}{c}, \quad dk = \frac{2\pi}{c} d\nu.$$

Then, the *number* of modes between k and $k + dk$ translates into the number of modes between ν and $\nu + d\nu$:

$$dN = \frac{dN}{dk} dk = \frac{dN}{dk} \frac{2\pi}{c} d\nu.$$

We differentiate $N_{\leq k} = \frac{L^3}{3\pi^2} k^3$:

$$\frac{dN}{dk} = \frac{L^3}{\pi^2} k^2.$$

Hence,

$$dN = \frac{L^3}{\pi^2} k^2 \times \frac{2\pi}{c} d\nu = \frac{2L^3 k^2 \pi}{\pi^2 c} d\nu.$$

Since $k = \frac{2\pi\nu}{c}$,

$$k^2 = \left(\frac{2\pi\nu}{c}\right)^2 = \frac{4\pi^2 \nu^2}{c^2}.$$

Then,

$$dN = \frac{2L^3}{\pi^2 c} \left(\frac{4\pi^2 \nu^2}{c^2}\right) \pi d\nu = \frac{8\pi L^3 \nu^2}{c^3} d\nu.$$

Therefore, the *density of modes* (i.e., number of modes per unit frequency) in the volume $V = L^3$ is

$$\frac{dN}{d\nu} = \frac{8\pi L^3 \nu^2}{c^3}.$$

3. Average Energy per Mode

In thermal equilibrium at temperature T , each electromagnetic mode of frequency ν has an average energy (from Planck's oscillator hypothesis) given by

$$\langle E \rangle_\nu = \frac{h \nu}{e^{\frac{h \nu}{k_B T}} - 1}.$$

This result can be derived using the canonical ensemble for a harmonic oscillator with quantized energy levels $E_n = n h \nu$ and the Boltzmann factor $\exp(-E_n/k_B T)$.

4. Energy Density

4.1. Total Energy in the Cavity

The total energy in all modes between ν and $\nu + d\nu$ is

$$dU = (\text{number of modes}) \times (\text{energy per mode}) = \frac{dN}{d\nu} \langle E \rangle_\nu d\nu.$$

Thus,

$$dU = \left[\frac{8\pi L^3 \nu^2}{c^3} \right] \left[\frac{h \nu}{e^{\frac{h \nu}{k_B T}} - 1} \right] d\nu.$$

4.2. Energy Density per Unit Volume

Since the volume of the cavity is $V = L^3$, the *energy density* $u(\nu)$ (energy per unit volume) in the frequency range ν to $\nu + d\nu$ is

$$u(\nu) d\nu = \frac{1}{V} dU = \frac{8\pi \nu^2}{c^3} \frac{h \nu}{e^{\frac{h \nu}{k_B T}} - 1} d\nu.$$

Often written more compactly:

$$u(\nu) d\nu = \frac{8\pi h \nu^3}{c^3} \frac{1}{e^{\frac{h \nu}{k_B T}} - 1} d\nu.$$

This is the well-known **Planck radiation law** for the spectral energy density of black-body radiation.